

Paola Romero

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EDUCATION

Massachusetts Institute of Technology (MIT)

Class of 2027

Cambridge, MA

- Candidate for B.S. in Mechanical Engineering. GPA: 4.7/5.0
- Involvements: Pappalardo Lab Apprentice, Edgerton Center Student Project Lab Asst. Shop Manager, Electronics for Mechanical Engineers Lab Assistant, Climbing Club, Wind Ensemble, Spinning Arts club
- Relevant Courses: Product Design, Design and Manufacturing I & II, Dynamics and Controls I & II, Thermal-Fluids Engineering I & II, Mechanics and Materials I & II, Numerical Computation, Differential Equations

EXPERIENCE

Lexington Medical

May. 2026-Aug. 2026

Cambridge, MA

Robotic Surgical Stapling - R&D Mechanical Engineer

- Characterized mechanical properties of strings/wires through ~300 loading/unloading cycles and 37 break tests
- Developed a test fixture for a 2-DOF mechanism; results informed motor requirements and articulation performance.
- Designed a 4-channel sensor testing PCB with Wheatstone bridges, amplification, temperature compensation, filtering, and power regulation.

Fabrication Integrated Design Lab (FIDL)

Sep. 2025-May. 2026

Cambridge, MA

Soft Robotics - Undergraduate Researcher

- Designed a bidirectional, tendon-driven grasper with both sequential and linked-joint actuation, enabling complex grasping tasks.
- Developed a model using pseudo-rigid body modeling to characterize the grasper's bending.
- Awarded 3rd Place for Undergraduate Science at 2026 MIT MechE De Florez Competition, presented at 2026 New England Manipulation Symposium, and currently preparing for RoboSoft 2027 conference paper submission.

Center of Innovation and Manufacturing at UPC

Jun. 2025-Aug. 2025

Barcelona, Spain

Additive Manufacturing - Research Intern

- Designed and prototyped a 6 m scale model of a 24 m post-tensioned concrete bridge in SolidWorks, including FEA simulations and hand calcs to validate performance under a 200 kN tensioning force and live loads.
- Created a novel modular mating system to join four 1.5m bridge sections

Campos Engineering

Jun. 2024-Aug. 2024

Dallas, TX

HVAC - Innovation Intern

- Assembled and tested dozens of SMT PCBs, cutting debugging time by 52% and improving quality control.
- Designed "Tulip," an electronics enclosure for an air velocity grid, improving airflow and reducing assembly time.
- Built a custom MIG-welded steel lift for an air velocity grid, rated for 40 kg load, ensuring durability and safety.
- Created the gateway PCB in KiCad—schematic, layout, routing, and component selection for signal reliability.

Campos Engineering

Jun. 2023-Jul. 2023

Dallas, TX

Innovation Intern

- Programmed Digi XBee 3 modules in MicroPython for temperature/humidity monitoring; soldered 30+ sensor PCBs and identified HVAC malfunctions in Campos Engineering's Dallas office.
- Iterated on 1mm wide airtight seals for airflow instruments, reducing manufacturing cost by 95%.
- Designed a 3D-printed wireless pressure sensor case, cutting sensor cost by 68% and streamlining assembly.
- Developed 3 unique prototypes for custom tools and components.

SKILLS

- **Technical:** Additive Manufacturing (SLA & FDM), Arduino & Raspberry Pi Microcontrollers, CNC Machining (Mill), Manual Lathe, MIG Welding, Injection Molding, Sheet forming (Metal and Thermoplastic), and Power Tools
- **Software:** CAD (SolidWorks, Fusion 360, Onshape), FEA (SolidWorks), CAM (Fusion360), MATLAB, Arduino (C++), KiCad, Python, HTML, CSS, Javascript